

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION
SSC PART-I (CLASS 9) PRACTICE EXAMINATION
CHEMISTRY — PAPER 23 (Version B)

Syllabus: Chapter 10 – Acids, Bases and Salts (35 marks) + Revision: Chapters 4, 6, 8, 9 (15 marks)

Total Marks: 50

Time Allowed: 2 hours 15 minutes

Version B

HIDAYAAT (Instructions)

1. **Saaray sawal** hal karnay hain. Is paper mein koi bhi **optional sawal nahin** hai.
2. Jawab apni notebook mein likhein. Har sawal ka number **bilkul waisa hi** likhein jaisa paper par hai, aur tarteeb se hal karein.
3. Section A mein sawal ka number likh kar sirf apnay option ka **letter** likhein, jaise **1. (b)**. Check kar lein ke jo letter aap ne likha hai, wohi option aap ne choose kiya hai.
4. Har calculation ka **working zaroor dikhayein**. Working theek ho to final jawab ghalat honay par bhi marks miltay hain.
5. **Calculator use kar sakti hain**. Har numerical jawab ke saath **unit** zaroor likhein.
6. Jahan sawal mein **state symbols** maangay gaye hain, wahan **(s), (l), (g), (aq)** likhna zaroori hai. Sirf balanced equation par poore marks nahin milein ge.
7. Har explanation aur calculation ke aakhir mein **ek mukammal jumla** likhein jo sawal ka jawab de.

SECTION A — MULTIPLE CHOICE QUESTIONS (10 × 1 = 10 MARKS)

#	Question	A	B	C	D	Answer
1	In the reaction $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$, the water molecule is behaving as a	Bronsted base	neutral solvent	Bronsted acid	a catalyst	(A)(B)(C)(D)
2	Which one of the following hydroxides is a base but not an alkali?	NaOH	$\text{Cu}(\text{OH})_2$	KOH	$\text{Ca}(\text{OH})_2$	(A)(B)(C)(D)
3	Which statement correctly describes a strong acid?	It dissociates completely into ions in aqueous solution	It dissolves completely in water	It has more hydrogen atoms in its formula	It changes red litmus to blue	(A)(B)(C)(D)
4	Dilute sulphuric acid is poured on a piece of marble, CaCO_3 . The gas that bubbles off is	hydrogen	ammonia	oxygen	carbon dioxide	(A)(B)(C)(D)
5	Ammonium chloride is warmed with sodium hydroxide solution. The gas given off is	ammonia	hydrogen	chlorine	carbon dioxide	(A)(B)(C)(D)
6	The aqueous solution of every acid contains an excess of	OH^- ions	H^+ ions	metal ions	oxide ions	(A)(B)(C)(D)
7	Even in a clean area, ordinary rain is slightly acidic. This is caused by dissolved	sulphur dioxide	nitrogen dioxide	carbon dioxide	oxygen	(A)(B)(C)(D)
8	In the manufacture of margarine, vegetable oils are hydrogenated using a catalyst of	iron	platinum	palladium	nickel	(A)(B)(C)(D)
9	An element X has the electronic configuration 2, 8, 2. X belongs to	Group 2, Period 3	Group 8, Period 3	Group 12, Period 3	Group 2, Period 2	(A)(B)(C)(D)
10	A reversible reaction in a closed container has reached dynamic equilibrium. At that moment the rate of the forward reaction is	greater than the rate of the reverse reaction	equal to the rate of the reverse reaction	zero	smaller than the rate of the reverse reaction	(A)(B)(C)(D)

SECTION B — SHORT ANSWER QUESTIONS (6 × 4 = 24 MARKS)

Chhay ke chhay sawal hal karein. Har sawal 4 marks ka hai.

2.	(i) Define a Bronsted-Lowry acid and a Bronsted-Lowry base. (2) (ii) In the reaction $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$, state which substance is the Bronsted-Lowry acid and	4
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	which is the Bronsted-Lowry base, and give a reason for each answer. (2)	
3.	Write the equation for the ionization of each of the following in water, using a single arrow for complete ionization and a reversible arrow for partial ionization: (i) HNO_3 (ii) H_2SO_4 (iii) CH_3COOH (3) (iv) Which one of the three is a weak acid, and why? (1)	4
4.	Write a balanced chemical equation, including state symbols , for each reaction below, and name the gas that is released in each case. (i) zinc with dilute hydrochloric acid (2) (ii) solid sodium carbonate with dilute hydrochloric acid (2)	4
5.	(i) All alkalis are bases, but all bases are not alkalis. Explain this statement. (2) (ii) Give one example of a base that is an alkali and one example of a base that is not an alkali, writing the formula of each. (2)	4
6.	(i) State two characteristic properties of bases, other than their taste. (2) (ii) Bases decompose ammonium salts on heating. Write the equation for the reaction of potassium hydroxide with ammonium chloride and name the gas liberated. (2)	4
7.	A student needs 0.2 mol of sodium hydroxide, NaOH , to prepare a solution. (i) Calculate the molar mass of NaOH . (1) (ii) Calculate the mass of NaOH she must weigh out. (2) (iii) Calculate the number of formula units present in this amount. (1) ($\text{Na} = 23$, $\text{O} = 16$, $\text{H} = 1$, $N_A = 6.02 \times 10^{23}$)	4

SECTION C — LONG ANSWER QUESTIONS (2 × 8 = 16 MARKS)

Dono sawal hal karein. Har sawal 8 marks ka hai.

8.	Oxides, hydroxides and neutralization. (i) Bases are the oxides and hydroxides of metals. Write a balanced equation for the burning of (a) sodium and (b) magnesium in oxygen, and name the oxide formed in each case. (3) (ii) Metal oxides react with water to form metal hydroxides. Write a balanced equation for the reaction of water with (a) Na_2O , (b) CaO and (c) MgO , naming the hydroxide formed in each case. (3) (iii) Write the general word equation for a neutralization reaction, give one balanced example from the chapter, and state what is always formed. (2)	8
9.	Energy changes in a reaction. (i) Define activation energy. (2) (ii) Draw an energy-profile diagram for an exothermic reaction, clearly labelling the reactants, the products, the activation energy E_a and the enthalpy change ΔH . (4) (iii) On the same diagram, sketch the pathway that is followed when a catalyst is added. State clearly what happens to E_a and what happens to ΔH . (2)	8

————— END OF QUESTION PAPER —————

Chapters covered: Chapter 10 Acids, Bases and Salts (Q1–Q7 of Section A, and Questions 2 to 6 and 8) | Revision: Chapter 4 Periodic Table, Chapter 6 Stoichiometry, Chapter 8 Energetics, Chapter 9 Chemical Equilibrium.

Student's Name: _____

Date: _____

Marks Obtained: _____ / 50